25901225 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCECT201 - AI FOR SMART ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Identify the primary function of a virtual assistant. [U] / [CO1]

   a). Data storage b). Hardware control

   c). User interaction and task assistance d). Network management

2. Define unsupervised learning. [K] / [CO2]

   a). Learning with labeled data b). Learning without data

   c). Learning without labeled data d). Learning using rewards

3. Identify from option,the storage requirement is measured in [K] / [CO3]

   a). Hertz b). Volts

   c). Bytes d). Pixels

4. Select the primary goal of mathematical optimization in AI. [U] / [CO1]

   a). Human imitation b). Knowledge representation

c). Finding the best solution under constraints

d). Natural language understanding

5. Name the basic unit of a neural network. [K] / [CO2]

a). Node

b). Neuron

c). Layer

d). Weight

6. Why are face apps and customized object detection considered basic real-world applications of computer vision? [An] / [CO3]

a). They do not use algorithms

b). They work without sensors

c). They use only text data

d). They apply visual analysis to solve practical problems

7. In navigation applications, the "Shortest Path Problem" refers to: [U] / [CO1]

a). Finding the road with the most scenery

b). Minimizing a cost function (time or distance) between two nodes

c). Identifying all possible roads in a city

d). Maximizing the number of turns in a route

8. Select the learning type that combines labeled and unlabeled data. [U] / [CO2]

a). Supervised

b). Unsupervised

c). Reinforcement

d). Semi-supervised

9. Select the correct CAD applications requirement [U] / [CO3]

a). Low RAM

b). High GPU

c). Basic CPU

d). Low storage

10. Name the process of improving model accuracy using past errors. [K] / [CO2]

a). Testing

b). Validation

c). Training

d). Deployment

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Identify one limitation that both Narrow AI and Generative AI share, which General AI would theoretically overcome. [U] / [CO1]

12. Define classification and regression. [K] / [CO2]

13. Why is camera resolution an important factor while selecting hardware for computer vision. [An] / [CO3]

14. Classify the main approaches to simulating the human brain, such as bottom-up and top-down methods. [U] / [CO1]

15. List the main components of a reinforcement learning system. [K] / [CO2]

16. Show how AI upscaling can improve user experience in streaming services. [U] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Compare Artificial Intelligence with human intelligence in terms of learning and reasoning. [U] / [CO1]

18. Discuss the role of neural network architecture in smart electronic systems. [U] / [CO2]

19. Draw and label a simple deep neural network architecture. [K] / [CO2]

20. Illustrate the working of Convolutional Neural Networks (CNNs) in computer vision applications with a neat diagram. [An] / [CO3]

[10x2 = 20]